Oluwatosin Ayeni

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Fractile

Dear Hiring Team,

I am writing to apply for the ML Runtime Engineer role at Fractile. I have deep hands-on experience integrating vLLM and SGLang inference engines, implementing and profiling KV cache strategies, and characterising paged attention behaviour at GPU level — which makes Fractile's mission of co-designing a hardware-native ML runtime with custom AI accelerators a natural next step for me.

Three specific achievements relevant to this role:

- **vLLM and SGLang runtime characterisation:** I benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× NVIDIA H100 NVL 94 GB (NVLink 4.0, CUDA 13.4), Qwen3-8B BF16, TP=2. Characterised paged attention and KV cache scheduling under memory pressure at 131K-token context: RadixAttention prefix caching delivered a 1,443× TTFT speedup (39 ms vs 55,565 ms), with 100% SLO compliance at concurrency 1–128. Live: inference-web-theta.vercel.app/benchmarks/llm
- **CUDA KV cache kernel profiling:** Using NVIDIA Nsight Systems and Compute, I profiled CUDA KV cache operations on RTX 4090 hardware, identified memory bandwidth as the primary bottleneck, and retuned kernel access patterns for a 3.8× throughput improvement — and 18× latency reduction via continuous batching redesign. Documented in phase2-llm-inference-deep-dive.
- **Speculative decoding and quantisation:** Ran speculative decoding experiments across vLLM and SGLang; implemented AWQ/GPTQ pipelines; built model architecture labs (transformer internals, GQA/MQA/MLA, VRAM modelling).

Fractile is building something genuinely novel: a hardware accelerator designed from the ground up for LLM inference, with a software runtime that can fully exploit its architecture. The ML Runtime Engineer role — integrating with vLLM and SGLang, building proof-of-concept KV cache management, and working on the paged attention layer — maps directly onto where I have spent the last two years. I'm happy to relocate to London or Bristol.

Yours sincerely,

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